Two Sum with a Hash Map

Topic: CompetitiveProgramming File: c1.pdf

PROBLEM

Given an array of integers and target T, return two different indices whose values sum to T, if such a pair exists.

APPROACH

Scan values from left to right. For value x, look for T-x in a hash map of previously seen values. If found, return the stored index and current index; otherwise store x.

EXAMPLE

A = [2,7,11,15], T = 9. When x=7, the map already contains 2, so return indices 0 and 1.

COMPLEXITY

Expected time: O(n) with a hash map. Extra space: O(n). Hash-table worst cases can differ from expected behavior.

EDGE CASES

Check duplicates, negative integers, multiple answers, and whether the problem expects indices or values.